EXPERIMENT - I

Linux Commands

Aim

To study various network commands used in linux and windows

1. **CODE**

arp -a

Output

(172.16.11.109) at 00:27:0e:13:f7:8c [ether] on enp350
(172.16.8.149) at 00.27:0e:13:f3:fe [ether] on enp350
(172.16.8.186) at 7C:57:58:38:C6:24 [ether] on enp350
(172.16.9.212) at 7C:57:58:38:da:93 [ether] on enp350

2. **CODE**

hostname

Output

localhost.localdomain

3. **CODE**

netstat -r

OUTPUT:

kernal IP routing table

Destination	Gateway	Gen mask	Flags	mssy window	irtt	Iface
default	gateway	0.0.0.0	UG	00	0	enP30
172.16.8.0	0.0.0.0	255.255.252.00	U	00	0	enp380

4. CODE

nslookup www.facebook.com

Output

Server : 172.16.8.1#

Address: 172.16.8.1#53

Non-authoritative answer

www.facebook.com

Canonical name = star-mini.c10r.facebook.com

Name: star-mini.c10r.facebook.com

Address : 57.144.56.1

5. CODE

route

Output

kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
default	gateway	0.0.0.0	UG	100	0	0	enp3s
172.16.8.0	0.0.0.0	255.255.252.0	U	100	0	0	enp3s

6. CODE

ip address show

Output

1: lo: inet 127.0.0.1/8 scope host lo

2: enp3s0: inet 172.16.8.127/22 brd 172.16.11.255 scope global enp3s0

7. ifconfig

Output:

enp3s0: flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST>
mtu 1500
inet 172.16.11.207
netmask 255.255.252.0
broadcast 172.16.8.255

8. mtr - serves as a network diagonistic tool

Output:

mtr google.com

localhost : local domain (1

Host
Packets

	loss %	snt	last	Avg	Best	Wrst	StDev
0.0	0%	17.8	0.1	0.1	0.0	0%	0.0

9. tcpdump - designed to capture and display packets

Output:

tcpdump -D

1. enp3s0 (up, Running)
2. any (Pseudo: device that captures on all interfaces) (up, Running)
3. lo (up, Running, Loopback)
4. wlp3s0
5. bluetooth0 (Bluetooth adapter number 0)

10. nbt stat - s

Output

NetBIOS names Resolution and Registration Statistics

Resolved by Broadcast = 0

Resolved by Name server = 0

Registered by Broadcast = 27

Registered by name server = 0

11. Path Ping -4 172.16.97.153

Output

over a max. of 30 hops

0 localhost [172.16.97.153]

1 localhost [172.16.97.153]

Computing statistics of 25 seconds

HOP	RTT	Source TO Here This node/Link Host/Sent = Pct	Host/Sent = Pct	Address
0			0/100 = 0%	172.16.7
1	0 ms	0/1000 = 0%	0/100 = 0%	172.16.7.15

Trace complete

Result

Thus the network command using linux and windows are successfully executed and the output is verified